



# Yeasin Bhuiyan

EEE Researcher & Embedded Systems Engineer

## Profile

Dedicated Electrical and Electronic Engineering (EEE) student at Jamalpur Science and Technology University (JSTU) specializing in practical circuit design, microcontroller systems, biomedical instrumentation, and autonomous hardware. Experienced in engineering real-time biopotential acquisition stations (ECG), developing high-speed multithreaded Python telemetry GUIs, designing autonomous fire-fighting robotics, and architecting IoT sensor simulations. Passionate about analog-to-digital interfacing and building reliable, real-world hardware prototypes.

## Project & Engineering Experience

2024 — Present | Jamalpur Science & Technology University

### Lead Hardware & Signal Developer — DIY ECG Machine

- Engineered a real-time Electrocardiogram (ECG) monitor integrating AD8232 biopotential analog front-end and Arduino Nano with serial synchronization.
- Designed multi-stage active analog filtration and instrumentation amplification to suppress 50Hz AC mains noise and baseline wander.
- Built a 60fps multithreaded Python GUI (Robofusion 1.0) with real-time waveform plotting and R-wave BPM calculation.
- Repository: [github.com/smsshaiful/Project-ECG-Machine-Arduino-Nano](https://github.com/smsshaiful/Project-ECG-Machine-Arduino-Nano) | Demo: [youtu.be/joZa\\_zyXXQM](https://youtu.be/joZa_zyXXQM)

2024 | Academic Hardware & Robotics Labs, JSTU

### Autonomous Robotics Developer — Fire Fighting Robot

- Developed an autonomous emergency response robot equipped with a 3-channel flame sensor array and L298N dual H-bridge motor driver.
- Implemented intelligent flame tracking algorithms and automated water pump servo targeting for rapid fire extinguishing.
- Demonstrated reliable fire suppression performance in physical test scenarios. Video Demo: [facebook.com/share/v/1BFJxyfnTD/](https://facebook.com/share/v/1BFJxyfnTD/)

2023 — 2024 | Autodesk Tinkercad Simulation & Prototyping

### Simulation & IoT Designer — Smart Home Automation

- Architected and simulated a multi-sensor smart-home control system in Tinkercad utilizing an Arduino Uno controller.
- Interfaced PIR motion detection for automated lighting, ultrasonic sensors for security monitoring, and gas leakage alert with buzzer actuation.
- Simulated relay-driven AC appliance control, optimizing power efficiency and component protection.

## References

### Department of EEE

Jamalpur Science & Technology University (JSTU)  
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### Academic References

Available Upon Request  
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Portfolio: <http://localhost:5500>

## CONTACT

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[github.com/smsshaiful](https://github.com/smsshaiful)

## EDUCATION

### B.Sc. in Electrical & Electronic Eng.

Jamalpur Sci. & Tech. Univ. (JSTU)  
Ongoing Undergraduate Program

### Higher Secondary Certificate (HSC)

Birshrestha Noor Mohammad Public College  
Completed: 2022 | Dhaka Board | GPA 5.00

### Secondary School Certificate (SSC)

Ragdail I.M. High School  
Completed: 2020 | Cumilla Board | GPA 5.00

## TECHNICAL SKILLS

- Circuit Design & Hardware Prototyping
- Digital Electronics & Logic Systems
- OrCAD / PSpice Circuit Simulation
- MATLAB & Simulink Modeling
- Python & C Programming
- Arduino Microcontrollers (Nano, Uno)
- Biomedical Instrumentation (ECG, AD8232)
- Sensor Interfacing & Telemetry
- Motor Drivers & Actuation (L298N)
- Tinkercad IoT Simulation

## PROJECTS & HONORS

### ROBOFUSION 1.0 (Clever Sapiens)

Real-Time ECG Monitor Demonstration

### Fire Fighting Robot Automation

Autonomous Sensor-Guided Suppression

### Smart Home IoT Simulation

Sensor & Relay Architecture in Tinkercad